MIL-C-1410F
30 September 1985
SUPERSEDING
MIL-C-1410E
20 February 1975

MILITARY SPECIFICATION

CHAIN, HOOK AND RING ASSEMBLY, TENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of chain, hook, and supporting ring for M-1942 assembly tent.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-C-271	- Chains and Attachments, Welded and Weldless
TT-E-485	- Enamel, Semigloss, Rust-Inhibiting
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD form 1426) appearing at the end of this document, or by letter.

FSC 8340

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FOLLOWING:

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-T-12879 - Treatment, Chemical, Prepaint and Corrosion Inhibitive, for Zinc Surfaces

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

4-1-107 - Chain, Hook and Ring, Tent Assembly

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 108 - Steel Bar, Carbon, Cold Finished (Standard Quality)
A 153 - Zinc Coating (Hot Dip) on Iron and Steel Hardware
B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 1005 - Measurement of Dry Film Thickness of Organic Coatings
D 2092 - Preparation of Zinc-Coated Steel Surfaces for Painting
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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at the end of this document | [illegible] Proposal (DD Form 1426) appearing

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

* 3.2 Materials. The materials used shall conform to the applicable referenced documents specified herein. Where materials are not definitely specified, they shall be of the quality and type normally used by the manufacturer provided that the completed item complies with all provisions of this document. Recycled material may be used (see 6.4).

* 3.2.1 Steel bar, carbon. The carbon steel bar shall conform to round form, cold finished (CF), composition 1008-1020, as applicable, of ASTM A 108.

3.2.2 Chain. The chain shall conform to type I, grade C, class 5, style 2, 7/32-inch nominal size of RR-C-271, except that the number of links per foot may be 13 1/3 in lieu of 14.

3.2.3 Ring, terminal. The terminal ring shall be size 7/16 by 2 1/2 inches in accordance with type VI of RR-C-271, except that the ring may be welded in lieu of being forged. In lieu of being forged, the terminal ring may be fabricated of 17/32-inch diameter round steel bar with welded ends and heat treated, providing the terminal ring has a minimum yield strength of 70,000 psi (at 0.2 percent offset) as specified in RR-C-271.

3.2.4 Enamel. Enamel shall conform to any type, composition G or L, olive drab color No. 24087 of TT-E-485.

3.2.5 Alternate components. Components offered as equivalent to components specified herein and on the referenced drawings as specific manufacturer's part number or equal shall be functionally equal to, of equal or better quality with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer.

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Prior to manufacture of the preproduction first article, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of first article or production manufacture as applicable and does not relieve the supplier of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.3 Construction. The construction of the chain, hook and ring assembly shall be in accordance with Drawing 4-1-107 and as specified herein. The end link of each chain shall be spread to allow movement of the chain without binding on the hook.

3.3.1 Link, terminal. The terminal link shall be fabricated of composition 1008-1020 steel as specified in 3.2.1 and ASTM A 108.

3.3.2 Ring, supporting. The supporting ring shall be fabricated of composition 1015-1020 steel as specified in 3.2.1 and ASTM A 108.

3.3.3 Welding. All welding shall be complete, sound, smooth, uniform, and free from burn through area, pits, slag inclusion, porosity, crack, and excess flux deposit.

3.4 Finish. All burrs, fins, rough edges, and sharp edges shall be ground smooth or removed prior to finishing.

* 3.4.1 Zinc coating. After complete fabrication, the chain, hook and ring assembly shall be zinc coated by any one of the following methods:

a. Electrodeposited zinc coating with supplementary phosphate treatment conforming to type IV, classification Fe/Zn13 Sc3, of ASTM B 633.

b. Hot dip zinc coating conforming to ASTM A 153 with preparation for painting in accordance with method A-crystalline zinc phosphate treatment of ASTM D 2092.

c. Commercial hot dip zinc coating with chemical treatment conforming to type I, class 1 or 2 of MIL-T-12879.

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3.4.2 Enameling. The zinc coated chain, hook and ring assembly shall be coated with enamel specified in 3.2.4. The finished coating shall be dry and free of any area of thin or no film, scratch, scuff, or abrasion exposing bare metal, blister, peel, soft run, dirt, or foreign matter embedded in finish, and wet or tacky paint.

* 3.4.3 Epoxy coating. In lieu of the zinc coating specified in 3.4.1 and enameling as specified in 3.4.2, the prepared surfaces shall be prepared and coated as follows:

a. The chain, hook and ring assembly shall be prepared by mechanical means. The surfaces shall be thoroughly and uniformly textured by sand blasting with 80 grit silicate. Prepared surfaces shall be kept clean during subsequent handling. Handling shall be by clean-gloved hands or by clean implements. The surfaces shall be retained in a clean environment. The surface shall be coated with epoxy not more than 8 hours after sand blasting.

b. The finish coating shall be green, color No. 24087 of FED-STD-595, electrodeposited epoxy powder coating conforming to Corvell ECA-1555-FC3 epoxy coating or its equal. Prior to use of the item of equal quality, the contractor shall submit the item to the contracting officer with supporting data for approval.

c. The electrodeposited epoxy powder coating shall be baked in accordance with commercial practice.

d. The thickness of the baked coating shall be a minimum of 0.003 inch when tested as specified in 4.4.3.3.

e. The final coating shall be smooth and uniform in appearance, free from wrinkles and streaks.

3.5 Workmanship. All components of the chain, hook and ring assembly shall be clean and free from dirt, grease, split, fracture, burrs, fins, rough edges, and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), the chain, hook and ring assembly shall be examined in accordance with table II and 4.4.3.2. The presence of any visual or dimensional defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase documents.

4.4.2 In-process inspection.

4.4.2.1 Process examination. Examination shall be made during the manufacturing process for the requirements listed in table I to establish that no deviation is made from the indicated requirements. Whenever a nonconformance is noted, correction shall be made to the affected items and the process.

TABLE I. Process examination

Requirement or operation	Requirement paragraph
Welds are complete, sound, smooth, uniform, and free from burn through area, pits, slag inclusion, porosity, crack, and excess flux deposit	3.3.3
Chain, hook and ring assembly is zinc coated with applicable treatment prior to enameling	3.4.1

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4.4.3 End item inspection. The inspection lot shall consist of all the chain, hook and ring assemblies offered for inspection at one time. The sample unit for this examination shall be one completely fabricated chain, hook and ring assembly.

4.4.3.1 Visual examination. Visual examination of the chain, hook and ring assemblies shall be made for the defects listed in table II. The inspection level shall be II and the acceptable quality level (AQL) shall be 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Category	
		Major	Minor
Finish	Not finished	101	
	Color of finish is not as specified		201
	Area of no film:		
	- Bare metal showing in area of 1/2 square inch or more	102	
	- Bare metal showing in area of less than 1/2 square inch		202
	Area of thin film:		
	- Zinc coating shows through in area of 1 square inch or more	103	
	- Zinc coating shows through in area of less than 1 square inch		203
	Scratch, scuff or abrasion exposing bare metal:		
	- 6 inches or more in longest dimension	104	
	- Less than 6 inches in longest dimension		204
	Blister, peel, soft run, dirt or foreign matter embedded in finish		205
	Wet or tacky paint		206

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TABLE II. Classification of defects - Continued

Examine	Defect	Category	
		Major	Minor
Construction and workmanship	Any component not fabricated of applicable referenced materials	105	
	Any component missing	106	
	Any component not fabricated as specified:		
	- End link of chain not spread and is binding to a degree that it will not allow movement of the end link of the chain on the hook	107	
	- End link of chain not spread to allow movement without binding of the end link on the hook, but will function		207
	Any component not clean (i.e., evidence of grease or dirt)		208
	Split or fracture	108	
	Burr, fin, rough edge or sharp edge		209
	More or less than the specified number of links per chain	109	
Welding	Missing	110	
	Not type of weld specified	111	
	Not complete or not sound	112	
	Has burn through area or crack	113	
	Not smooth, not uniform or has pits, slag inclusion, porosity, or excess flux deposit		210

4.4.3.2 Dimensional examination. Examination shall be made of the end item for defects in dimensions. Any dimension not within the specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

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* 4.4.3.3 Epoxy coating thickness examination (when applicable). The end item shall be examined for thickness of the epoxy coating in accordance with ASTM D 1005. Any epoxy coating thickness less than specified in 3.4.3 shall be classified as a defect. The lot size shall be expressed in units of complete assemblies. The sample unit shall be one assembly. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.4.4 Packaging inspection. An inspection shall be made to determine that preservation, packing, and markings are in compliance with section 5. Defects shall be as specified in table III. The lot shall be the number of shipping containers offered for inspection at one time. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

TABLE III. Packaging inspection

Examine	Defect
Markings (interior and exterior)	Missing; incorrect; incomplete; illegible; of improper size, location, sequence, or method of application
Materials	Any nonconforming component; component missing, damaged, or otherwise defective
Workmanship	Bulged or distorted container Inadequate application of components, such as: shipping container or case liner (when required) not as specified, incomplete closure of container, loose strapping, improper taping, or inadequate stapling
Contents	Quantity of chain, hook, and ring assemblies per unit pack or shipping container, as applicable, is more or less than required

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* 4.4.5 Palletization inspection. An inspection shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with table IV. The lot size shall be the number of palletized unit loads in the inspection lot. The sample unit shall be one palletized unit load fully packaged. The inspection level shall be S-1 and the AQL shall be 6.5, expressed in terms of defects per hundred units.

TABLE IV. Palletization inspection

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Three chain, hook and ring assemblies shall be arranged in the most compact manner with chains placed within the rings and unit packed in a snug-fitting fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic or weather-resistant of PPP-B-636. Closure shall be in accordance with the appendix of PPP-B-636.

* 5.1.2 Commercial. Chain, hook and ring assemblies shall be preserved in accordance with ASTM D 3951.

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* 5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Nine chain, hook and ring assemblies, preserved as specified in 5.1, shall be packed in a shipping container conforming to overseas type of PPP-B-601; or class 2, style 2 or 4 of PPP-B-621. Each shipping container shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547, except that when class weather-resistant interior boxes are used and each box is waterproofed in accordance with the appendix of PPP-B-636, case liners shall not be required. Closure and strapping shall be in accordance with the appendix of the applicable container document.

* 5.2.2 Level B packing. Three chain, hook and ring assemblies, preserved as specified in 5.1.1, shall require no further packing.

5.2.3 Commercial packing. Chain, hook and ring assemblies, preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), chain, hook and ring assemblies, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type I or Ia, as applicable, of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The chain, hook and ring assemblies are used in the erection of M-1942 assembly tents and to furnish means of supporting the roof of the tent at the peak.

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* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for inspection and approval of the first article.

* 6.4 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.2).

6.5 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue have been made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-1410F	Chain, Hook and Ring Assembly, Tent

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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